

Statistical analysis on points scored in the National Football League

Plan of Investigation:

Football has a major part in my life, ever since I was little I have been playing football with my family in the backyard, and studying the game. When I started high school, I took it to the next level by playing tackle football for the first time and I am loving it. Every Sunday, me and my family watch football, and pay close attention to our favorite teams even if they are not doing well in a particular year. Football has been there with me through the ups and downs of my life, and I know I can count on a good Sunday slate of games to cheer me up after a rough night on Friday.

To start, I will see if points scored (amount of points a team scores in one season) for NFL teams in 2019 is correlated to their win percentage. I will do this by using my data to compute a correlation coefficient r , and graph the data to visually see if they are correlated. If the correlation coefficient is greater than 0.7, then it would be considered strongly correlated. Then I would also calculate the least squares regression line for the data which I can use any number of points scored and can calculate the expected win percentage for that amount of points scored. Even though a team that scores more points *should* win does not always mean that they *will* win. For example, in 2019 the San Francisco 49ers and New Orleans Saints played and the score was a 48-46 49ers victory. Even though both teams scored a lot of points, one of them still lost.

Then I will use a two sample t-test to test if there is enough evidence to prove if there is a significant difference between the mean number of points scored in different decades. This will help answer my research question because it will show that NFL offenses are changing through each decade which shows that points scored has an effect on the National Football League.

Obtaining Data:

Since I wanted to compare NFL team statistics, I used data from Pro Football Reference¹ to get the most accurate data possible. Since there are only 32 NFL teams, I would simply extract each team's statistics and compare them to their win percentage. I obtained data from 2019 because in 2020 many players were fearful of the COVID-19 pandemic that could affect NFL players, many of them did not play for the 2020 season and still the 2021 season. Once I found the data that I wanted, I put them into a list on google sheets² which made the data easily readable and had the ability to create graphs as well. After I calculated each team's win percentage³ I had something to compare other data to to see if they were correlated. Still using Pro Football Reference, I found other data.

Table 1: Team, Win percentage and points for (2019)

Team	Points for	Win %
Arizona Cardinals	361.00	34.40%
Atlanta Falcons	381.00	43.75%
Baltimore Ravens	531.00	87.50%
Buffalo Bills	314.00	62.00%
Carolina Panthers	340.00	31.25%
Chicago Bears	280.00	50.00%

¹ "Pro-Football-Reference.com: Pro Football Statistics and History."
<https://www.pro-football-reference.com/>. Accessed 17 Jan. 2022.

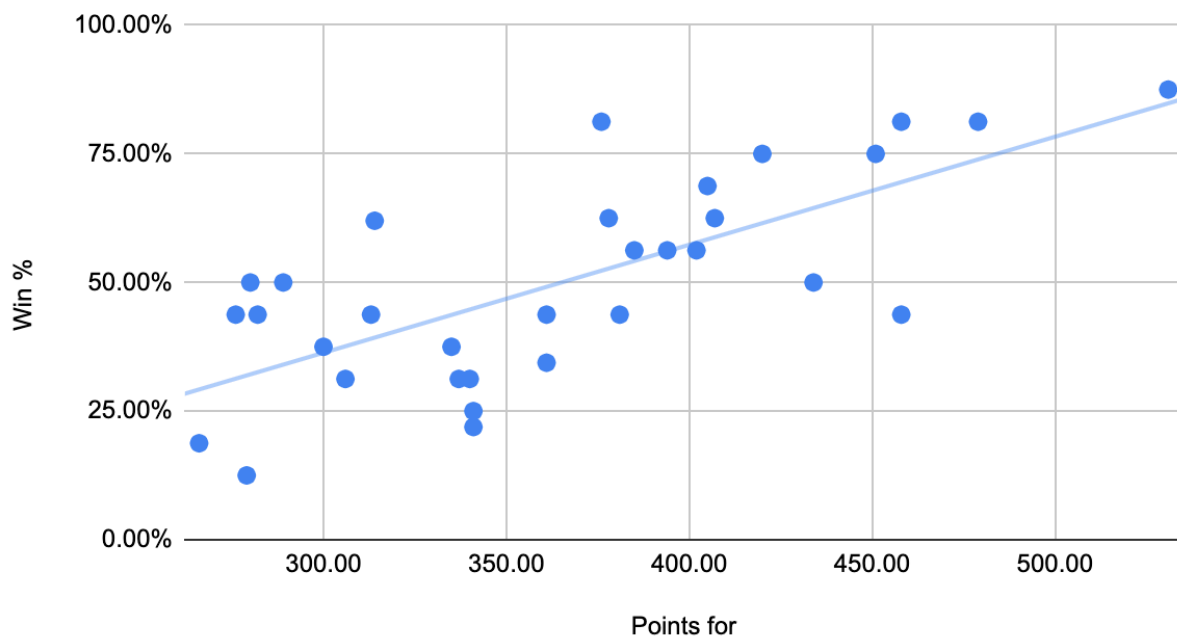
² "Google Sheets: Free Online Spreadsheet Editor." <https://www.google.com/sheets/about/>. Accessed 22 Jan. 2022.

³ "Winning Percentage Calculator." 12 Jan. 2022,
<https://www.omnicalculator.com/sports/winning-percentage>. Accessed 22 Jan. 2022.

Part 1: Points for Compared to Win Percentage Scatter Plot

After obtaining each team's win percentage and their respective total points for the 2019 season, I decided to graph my data in a scatter plot because it would allow me to see the correlation between these two data sets clearly. The points for would go on the x-axis because it is the independent variable, and the win % would go on the y-axis because it is the dependent variable. This is because I am trying to see if points for affects it.

Win % vs. Points for



After looking at the scatter plot, I noticed that it had a strong, positive, and linear correlation between points for and win percentage. The data is very close together, which made me curious as to what the Pearson's product moment correlation coefficient is. Using my TI-84 Plus CE calculator, I calculated the correlation coefficient of this data, (r) the least squares regression line, $(0.202x - 24.62)$ and the coefficient of determination value (r^2).

Pearson's Product Moment Correlation Coefficient:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

$$r = 0.688685308$$

The correlation coefficient shows the strength of the data; since the correlation coefficient is approximately 0.7. It shows that points for and win percentage for NFL teams is strongly correlated in 2019.

The Coefficient of Determination:

$$r^2 = 0.4742874535$$

The coefficient of determination, according to wikipedia⁴, “is the proportion of the variation in the dependent variable that is predictable from the independent variable(s).”. This means that 47% of points scored across the league can be explained by the variation of the win percentage. In contrast, 53% of the points scored across the league can be explained by other factors.

Least Squares Regression Line:

$$a = 0.2021798616 \text{ Win \%} = 0.202 \times \text{points for} - 24.6$$

$$b = -24.62221199$$

This is the line of best fit for the graph above and is shown as the light blue line. The line means that for every increase in points for, the win percentage increases by 0.2 percent.

⁴ "Coefficient of determination - Wikipedia." https://en.wikipedia.org/wiki/Coefficient_of_determination. Accessed 17 Jan. 2022.

Part 2: Finding the Expected Value

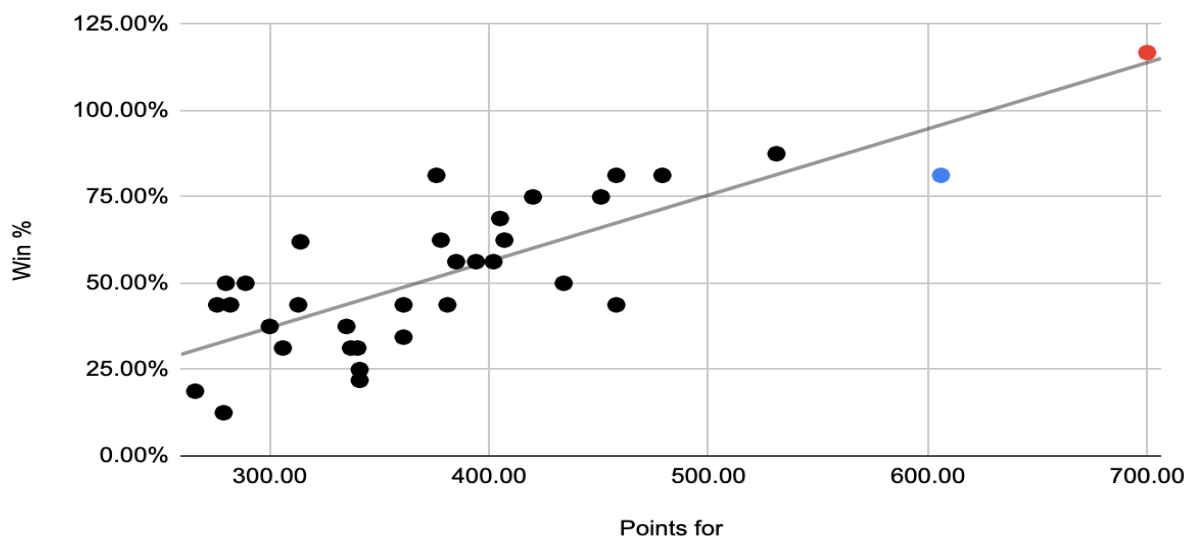
Using the least squares regression line above, I can now figure out what the expected win percentage for any amount of points for. For comparison, the 2013 Denver Broncos' points for was 606 and their win percentage was 81.25%. I decided to use 700 points because the league record for points scored in a single season is 606, and I was curious to see what win percentage I would expect for a team to get with more than this record. Then, I calculated the expected win percentage for a team who scored 700 points in one season.

$$\text{Win \%} = 0.202 \times 700 - 24.6$$

$$\text{Win Percentage} = 116.8$$

This value means that the team that scores 700 points in one season should win more than 100% of the time. This is improbable because you can't win more than 100% of your games. Using my calculations, I graphed the new data points on an updated graph. The blue dot represents the 2013 Denver Broncos record setting season and the red dot represents my created data point and calculations above.

Win % vs. Points for (with expected value)



As you can see the new points dramatically affect the scatter plot by increasing the strength of it since it is very close to the line of best fit, which is where I expected it to be. Since this is extrapolation (past the original domain and range), it would not be reliable to predict in a real life situation. Nonetheless, I was further curious to find the new r value.

Pearson's Product Moment Correlation Coefficient:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

$$r = 0.8008279733$$

The correlation coefficient shows the strength of the data; since the correlation coefficient is 0.8 it shows that the added data points increases the strength of the correlation between win percentage and points for. This is much more than the previous r value, which shows that the new expected point is so close in relation to the rest of the data that it increases the r value.

Comparing Mean Points Scored for Different Decades- Part 1

Now that I know that points for and win percentage are correlated, I was curious to see if there was a significant difference between the mean points scored for different decades. I will do this by conducting a t test with a 5% significance level. I used data from Pro Football Reference again because they have very accurate National Football League data going back before the super bowl era. Also I used 2019 data because it was the last “normal” year for the NFL since COVID-19 had not affected the player's decisions to play that year or not. I used data from 2009 because it is exactly 10 years before 2019. This way I can see if there is a difference between the two means. If there is a significant difference between them, then that could indicate if the NFL teams were changing strategies. For example, if they were figuring out new and more efficient

ways to pass the football that led to more points, it would show a significant difference and ultimately a new style of play.

Table 2: Points Scored in 2019 vs. Points Scored in 2009

2019 points scored	2009 points scored
361.00	375.00
381.00	363.00
531.00	391.00
314.00	258.00
340.00	315.00

Before I start, I will make sure both data sets are both normally distributed because in order to perform a T-test I need to ensure that this data is normally distributed. If the data is not normally distributed, then the data is skewed. To do this, I will do a chi squared goodness of fit test against a normal distribution for both 2019 and 2009. A goodness of fit test will compare the expected values to the observed values. If they match up well, then the data is considered normal. To get each expected value, I used the normalcdf function in my calculator and then multiplied by the amount of teams in the NFL at this time.

Chi Squared Goodness of Fit Test for points scored in 2019:

Observed values (2019)		Expected value (2019)	
Ranges	Frequencies	Ranges	Frequencies
$260 \leq x < 320$	10	$260 \leq x < 320$	6.162
$320 \leq x < 360$	5	$320 \leq x < 360$	6.978
$360 \leq x < 420$	10	$360 \leq x < 420$	10.307
$420 \leq x < 560$	8	$420 \leq x < 560$	5.226

I calculated the observed and expected values for 2019 (above) to conduct a 5% significance level Chi Squared Goodness of Fit Test.

Null Hypothesis: Points scored in 2019 follows a normal distribution.

Alternative Hypothesis: Points scored in 2019 do not follow a normal distribution.

Chi Squared Value: $\chi^2 = 4.432789872$

The chi squared value tells me the difference between the observed frequencies and the expected frequencies. Since this number is relatively low, then it means that there is not that much difference between the observed and expected values.

P-Value: $p = 0.2183641974$

The p-value tells me the probability that the observed data will fit the expected data.

Degrees of Freedom: Degrees of Freedom = 3

Degrees of freedom is the number of values that are able to vary when χ^2 is known. In this case, the degrees of freedom is 3 so 3 values are able to vary.

Conclusion: **Fail To Reject**

Since the p-value is greater than the significance level, then I fail to reject the null hypothesis.

This means that Points scored in 2019 follow a normal distribution and that the majority of the points scored in 2019 are similar with few outliers.

Chi Squared Goodness of Fit Test for points scored in 2009:

Observed values (2009)		Expected values (2009)	
Ranges	Frequencies	Ranges	Frequencies
$170 \leq x < 290$	8	$170 \leq x < 290$	7.611
$290 \leq x < 350$	8	$290 \leq x < 350$	8.918
$350 \leq x < 410$	10	$350 \leq x < 410$	8.4199
$410 \leq x < 530$	7	$410 \leq x < 530$	6.2158

Here are the results from a Chi Squared Goodness of Fit Test at a 5% significance level for points scored in 2009:

Null Hypothesis: Points scored in 2009 follows a normal distribution.

Alternative Hypothesis: Points scored in 2009 do not follow a normal distribution.

Chi Squared Value: $\chi^2 = 0.5098409901$

P-Value: $p = 0.9167241305$

Degrees of Freedom: Degrees of Freedom = 3

Conclusion: **Fail To Reject**

Since the p-value is greater than the significance level, then I fail to reject the null hypothesis.

This means that Points scored in 2009 follow a normal distribution.

Now that both data sets are normal, I can conduct a two sample t test at a 5% significance level and find out if there is a significant difference between the mean points scored in 2019 and in 2009.

Two sample t test for mean points scored in 2019 and 2009:

Null Hypothesis: $\mu_1 = \mu_2$

The null hypothesis in this scenario means that the mean number of points scored in 2019 is equal to the mean number of points scored in 2009.

Alternative hypothesis: $\mu_1 \neq \mu_2$

The alternative hypothesis in this case is that the mean number of points scored in 2019 is not equal to the mean number of points scored in 2009. In other words, there is a significant difference between the two means.

T-Value: $t=1.158763513$

The t-value indicates if there is a significant difference between the two data sets. Since this t-value is small, then it shows that the data is more similar than significantly different.

P-Value: $p=0.2509963593$

The p value is the probability that you would get this data set if the null hypothesis was true. So if the null hypothesis is true (mean of points scored in 2019 is equal to the mean number of points scored in 2009) then there would be about a 25% chance that you would get this data set.

Conclusion: Fail to Reject

I would fail to reject the null hypothesis because the p-value is greater than the significance level. In other terms, $0.251 > 0.05$. In context, there is not enough evidence to prove that there is a significant difference between the mean number of points scored in 2019 and the mean number of points scored in 2009.

In reflection, I thought that this conclusion was appropriate because from a fan standpoint, there weren't really many new offensive styles. This means that coaches in the NFL didn't really develop major new plays that would make a significant impact on the team's scoring.

Comparing Mean Points Scored for Different Decades- Part 2

Since the last t test showed no significant difference, I was curious if I went back in time another decade if there would be a significant difference. Again, if there is a significant difference between the two means, then it would show that the game of football was evolving because there would be more points scored. Since the last test was comparing 2019 to 2009, I was curious to see what 2019 and 1999 would compare like. I still used Pro Football Reference to acquire this data. In this data set, there was one less team in the league in 1999 because the Houston Texans was not created yet. Since I don't know if this data is normal, I will have to perform another Chi Squared Goodness of Fit test, then a T-test to compare the two datasets' means.

Table 3: Points scored in 2019 and points scored in 1999

2019 points scored	1999 points scored
279.00	322.00
280.00	324.00
282.00	326.00
300.00	338.00

Chi Squared Goodness of Fit Test for points scored in 2019:

Observed values (1999)		Expected values (1999)	
Ranges	Frequencies	Ranges	Frequencies
$210 \leq x < 300$	12	$210 \leq x < 300$	8.622
$300 \leq x < 330$	7	$300 \leq x < 330$	5.26
$330 \leq x < 360$	3	$330 \leq x < 360$	5.367
$360 \leq x < 530$	9	$360 \leq x < 530$	10.633

Here are the results from a Chi Squared Goodness of Fit Test at a 5% significance level for points scored in 1999:

Null Hypothesis: Points scored in 1999 follows a normal distribution.

Alternative Hypothesis: Points scored in 1999 does not follow a normal distribution.

Chi Squared Value: $\chi^2 = 3.19375887$

P-Value: $p = 0.3627052379$

Degrees of Freedom: Degrees of Freedom = 3

Conclusion: **Fail To Reject**

Since the p-value is greater than the significance level, then I fail to reject the null hypothesis.

This means that Points scored in 1999 follow a normal distribution.

Now I can conduct a two sample t test at a 5% significance level and find out if there is a significant difference between the mean points scored in 2019 and in 1999.

Two sample t test for mean points scored in 2019 and 1999:Null Hypothesis: $x_1 = x_2$ Alternative Hypothesis: $x_1 \neq x_2$ T-Value: $t=1.879798056$ P-Value: $p=0.0649135985$ **Conclusion: Fail to Reject**

Since $0.065 > 0.05$, there is not enough evidence to prove that there is a significant difference between the mean number of points scored in 2019 and the mean number of points scored in 1999.

When reflecting upon this conclusion, I was surprised because I thought that there would be a significant difference in the mean number of points scored in 2019 and 1999 because I thought the large amount of time between the two years would make a significant difference.

Comparing mean points scored for different decades-part 3

Since part 2 showed that there was no significant difference between the mean points scored for 2009 and 1999, I decided to compare the data from 2019 and 1989 to see if there is a significant difference between these two years. I still used Pro Football Reference to obtain my data, and will still use a t-test. Since there were only 28 teams (Baltimore Ravens, Jacksonville Jaguars, Carolina Panthers, and Houston Texans not created yet) in 1989, there are only 28 data points for

1989. Since I don't know if the points scored in 1989 are normal, I will have to perform another Chi Squared Goodness of Fit test, then a T-test to compare the two datasets' means.

Table 4: Points scored in 2009 and points scored in 1999

2019 points scored	1989 points scored
279.00	334
280.00	342
282.00	348
300.00	351

Chi Squared Goodness of Fit Test for points scored in 1989:

Observed values (1989)		Expected values (1989)	
Ranges	Frequencies	Ranges	Frequencies
$200 \leq x < 290$	7	$200 \leq x < 290$	6.611
$290 \leq x < 320$	5	$290 \leq x < 320$	5.169
$320 \leq x < 350$	5	$320 \leq x < 350$	5.612
$350 \leq x < 470$	11	$350 \leq x < 470$	9.9788

Here are the results from a Chi Squared Goodness of Fit Test at a 5% significance level for points scored in 1989:

Null Hypothesis: Points scored in 1989 follows a normal distribution.

Alternative Hypothesis: Points scored in 1989 do not follow a normal distribution.

Chi Squared Value: $\chi^2 = 0.1996610565$

P-Value: $p = 0.9776439963$

Degrees of Freedom: Degrees of Freedom = 3

Conclusion: **Fail To Reject**

Since the p-value is greater than the significance level, then I fail to reject the null hypothesis.

This means that Points scored in 1989 follow a normal distribution.

Now I can conduct a two sample t test at a 5% significance level and find out if there is a significant difference between the mean points scored in 2019 and in 1989.

Two sample t test for mean points scored in 2019 and 1989:

Null Hypothesis: $x_1 = x_2$

Alternative hypothesis: $x_1 \neq x_2$

T-Value: $t = 2.144324466$

P-Value: $p = 0.0362093362$

Conclusion: Reject

Since $0.036 < 0.05$, there is enough evidence to prove that there is a significant difference between the mean number of points scored in 2019 and the mean number of points scored in 1989.

I thought this conclusion rightfully displayed the change of NFL offense because there was a lot that changed between 1989 and 2019. The creation of the K-Gun offense (a fast-paced offense used by the Buffalo Bills) in the 90's boosted the Buffalo Bills' scoring to the most in the League, which in turn raised the mean points scored. This is why I was mainly surprised as to why points scored in 1999 and 2019 were not statistically different because of this new style of play which raised the amount of points scored in the 90's.

Conclusion:

In conclusion points for does have an effect on the National Football League to a great extent. In the significance test between the mean points scored in 2019 and 1989, it showed that there is a difference between the two means. This tells us that between 2019 and 1989 there were more points scored overall. I got a pearson's product moment correlation, or r , value of 0.688. This shows that win percentage and points scored in 2019 are moderately correlated. I can pair this conclusion with the significance test to conclude that teams who score the most points typically win more games, and the amount of points scored has increased significantly since 1989. This means that these two conclusions are connected because new concepts, ideas, innovations, and technology can increase a NFL teams' win percentage. This is supported in NFL history because in this New York Times article⁵It talks about how in 1990 the Buffalo Bills used the "K-Gun" offense. In this offense the Bills used the "shotgun" (where the quarterback stands 4 to 5 yards behind the center) formation, used a "no huddle" (where the offense does not huddle to call the play) system which was used from another team in 1988, and spread their receivers out more. This never before seen offense scored a league high 428 points and helped the Bills reach the Super Bowl in 1990. This shows how the fusion of NFL offenses help produce more points, and in my exploration, I proved when these changes occurred.

If I were to do this exploration again, I could have used more factors, for example if interceptions had a correlation on a team's win percentage; however, I thought all of my tests and calculations were necessary to make my conclusion. I could also have factored in the rule changes, for example if the addition of the pass interference penalty had an effect on a team. Another possible limitation would be that my samples were not random. Although if I were to do

⁵ "How Offense Took Over the N.F.L. - The New York Times." 19 Jan. 2019, <https://www.nytimes.com/2019/01/19/sports/nfl-offense-records.html>. Accessed 22 Jan. 2022.

random sampling, I would have a lot less data values, and that would make my findings less accurate.

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Appendix:

Table 1 continued:

Cincinnati Bengals	279.00	12.50%
Cleveland Browns	335.00	37.50%
Dallas Cowboys	434.00	50.00%
Denver Broncos	282.00	43.75%
Detroit Lions	341.00	21.90%
Green Bay Packers	376.00	81.25%
Houston Texans	378.00	62.50%
Indianapolis Colts	361.00	43.75%
Jacksonville Jaguars	300.00	37.50%
Kansas City Chiefs	451.00	75.00%
Los Angeles Chargers	337.00	31.25%
Los Angeles Rams	394.00	56.25%
Miami Dolphins	306.00	31.25%
Minnesota Vikings	407.00	62.50%
New England Patriots	420.00	75.00%
New Orleans Saints	458.00	81.25%
New York Giants	341.00	25.00%
New York Jets	276.00	43.75%
Oakland Raiders	313.00	43.75%

Philadelphia Eagles	385.00	56.25%
Pittsburgh Steelers	289.00	50.00%
San Francisco 49ers	479.00	81.25%
Seattle Seahawks	405.00	68.75%
Tampa Bay Buccaneers	458.00	43.75%
Tennessee Titans	402.00	56.25%
Washington Redskins	266.00	18.75%

Table 2 continued:

280.00	327.00
279.00	305.00
335.00	245.00
434.00	361.00
282.00	326.00
341.00	262.00
376.00	461.00
378.00	388.00
361.00	416.00
300.00	290.00
451.00	294.00
337.00	360.00

394.00	470.00
306.00	427.00
407.00	510.00
420.00	402.00
458.00	348.00
341.00	197.00
276.00	429.00
313.00	368.00
385.00	454.00
289.00	330.00
479.00	280.00
405.00	175.00
458.00	244.00
402.00	354.00
266.00	266.00

Table 3 continued:

314.00	352.00
335.00	357.00
340.00	390.00
341.00	390.00

361.00	392.00
361.00	396.00
376.00	399.00
381.00	421.00
434.00	526.00
451.00	443.00
458.00	423.00
531.00	320.00
479.00	317.00
458.00	314.00
420.00	308.00
407.00	299.00
405.00	299.00
402.00	295.00
394.00	285.00
385.00	283.00
378.00	272.00
341.00	272.00
337.00	270.00

313.00	269.00
306.00	260.00
289.00	245.00
276.00	217.00
266.00	

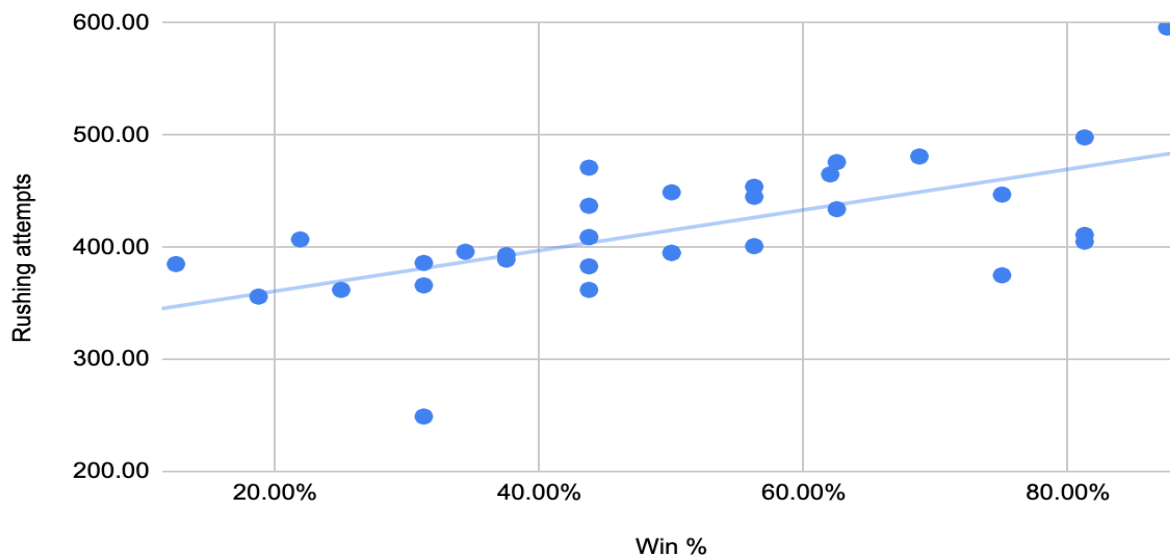
Table 4 continued:

314.00	358
335.00	362
340.00	362
341.00	365
361.00	386
361.00	386
376.00	404
381.00	409
434.00	426
451.00	442
458.00	331
531.00	320
479.00	318
458.00	315
420.00	312
407.00	298
405.00	297
402.00	279
394.00	266
385.00	265

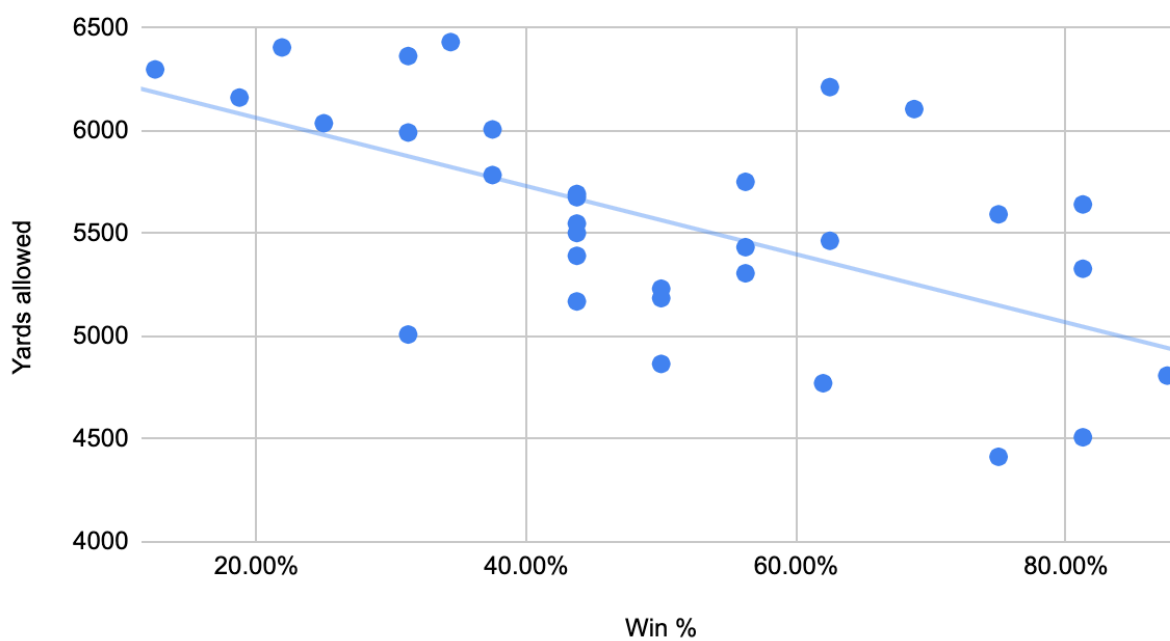
378.00	258
341.00	253
337.00	241
313.00	204
306.00	
289.00	
276.00	
266.00	

These are some other interesting statistics that I found that showed to have an effect on a team's

Rushing attempts vs. Win %

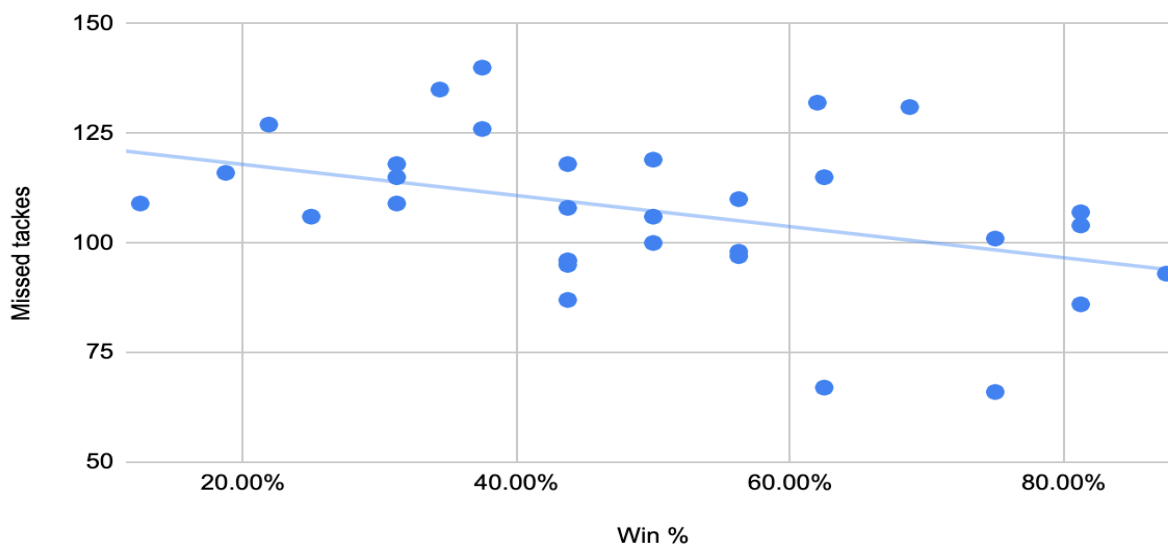


Yards allowed vs. Win %



win percentage.

Missed tackles vs. Win %



Win % vs. Average Time of Possession (excluding OT)

